Agentic AI: What Works, What Breaks, and What You Can't Trust

A peer-reviewed reality check for decision-makers

Section 6 lecture notes

Why another reality check, right now

Section 5 made a specific claim, grounded in the largest AI-researcher survey ever conducted [Grace et al., 2024]: expert timelines for human-level machine intelligence are unstable, narrow-task forecasts already undershot, and the tail of catastrophic outcomes is not a fringe position inside the research community. That lecture ended on a question: what do you, a manager, actually do about it.

Section 6 picks up where that question lands. The honest answer is not another strategic framework. The honest answer is that the category managers are being sold today, "agentic AI," behaves differently from the chatbots their pilots trained them to evaluate. It has scaffolding you can't see, benchmark scores you can't trust, and failure modes that are well-documented in peer-reviewed literature but have not yet made it into most procurement templates. This section unpacks each of those three, and ends with a short playbook you can take to Monday's meeting.

Everything below is anchored to papers published in venues where peer review is real: ICML, NeurIPS, ICLR, FAccT, and *Science* / *Patterns* (Cell Press). Where we cite a preprint or technical report, we flag it explicitly. Agentic AI is a fast-moving field, and a lot of the material that circulates in the trade press is not peer-reviewed. Managers who treat every widely-shared finding as settled science will be surprised, repeatedly.

1 What an agent actually is

The word "agent" has collapsed in meaning. In the research literature and, increasingly, in procurement documents, it refers to a loop that wraps a language model in three things: a *planner* that decides what to try, a set of *tools* (a browser, a code interpreter, a retrieval index, a company API), and a *reflection* mechanism that looks at the result and decides whether to retry.

A manager's first instinct is to ask which model is best. That is the wrong lever. The single most replicated finding in the agentic-AI literature is that the scaffolding, not the base model, accounts for most of the performance gain.

Yao et al. [2023] introduced ReAct (ICLR 2023), which interleaves a short reasoning step with a single tool action, then observes the result before planning the next step. On ALFWorld, a household-task benchmark, ReAct raised the same base model's success rate from roughly 34% to 71%. The base model did not change; the loop around it did.

Shinn et al. [2023] extended the idea in Reflexion (NeurIPS 2023). After a failure, the agent writes a paragraph in plain language about *why* it failed, stores that paragraph, and consults it on the

next attempt. This simple trick raised GPT-4's pass@1 on the HumanEval coding benchmark from roughly 80% to 91%.

Schick et al. [2023] made the sharpest point (NeurIPS 2023): a 6.7-billion-parameter model that teaches itself, in a self-supervised way, to call external APIs outperforms GPT-3 (175 billion parameters) on downstream tasks where tool use matters. Capability and model size are not the same lever. If you are evaluating vendors on raw model size, you are evaluating the wrong thing.

Park et al. [2023], at UIST 2023 (best-paper honorable mention), gave 25 language-model agents memory, reflection, and planning, then dropped them into a simulated town. Left alone, they organised a Valentine's Day party. The point is not that a simulated village is commercially important. The point is that the same scaffolding ideas, applied at population scale, produce emergent collective behaviour that no one explicitly programmed. Procurement conversations should ask what happens when *two* of your agents interact, not just when one does.

The procurement implication is the durable lesson from this block of research. You are not buying a model. You are buying a model plus planner plus tools plus retry loop plus guardrails plus integration. Vendor demos that highlight the model and hide the scaffolding are demonstrating the wrong thing.

2 What the benchmarks say, and what they hide

The agentic-AI literature settled on four benchmarks, all peer-reviewed at ICLR 2024: GAIA, SWE-bench, AgentBench, and WebArena. All four were designed to measure something closer to real work than prior LLM benchmarks like MMLU. All four are now being misquoted in vendor pitches. This section lays out what they actually say.

GAIA (Mialon et al., ICLR 2024)

GAIA is a set of 466 questions that a careful human can answer in minutes but that require an agent to plan, browse, reason over files, and use tools in combination. Questions are stratified into three levels of difficulty. Humans score about 92% overall. At release, the best LLM-plus-plugins configuration reached roughly 15% [Mialon et al., 2024].

As of April 2026, the top score on the public GAIA leaderboard is 44.8% (GPT-5 Mini). Humans remain at roughly 92%. Progress in two years: from 15% to 44.8%, a tripling. Gap remaining: 47 percentage points. Rate of closure: fast. Honest reading: capability has grown substantially; the gap is still the story.

SWE-bench (Jimenez et al., ICLR 2024, oral)

SWE-bench is a set of 2,294 real GitHub issues from popular open-source Python repositories, paired with the human-written patches that actually fixed them [Jimenez et al., 2024]. An agent is given the issue text and the repository, and must produce a patch that passes the project's own test suite. At release in October 2023, GPT-4 resolved roughly 2% of the issues.

As of April 2026, the top SWE-bench Verified score is 93.9% (Claude Mythos Preview). That is a forty-fold improvement in 30 months. But there is a catch. In mid-2024, OpenAI stopped reporting Verified scores for its own frontier models after detecting that every frontier system they tested showed signs of training-data contamination on the benchmark. This is documented on OpenAI's

own announcement page for the Verified subset [OpenAI, 2024], and echoed in independent audits such as the Berkeley RDI group's widely-read 2026 post "How We Broke Top AI Agent Benchmarks."

The managerial takeaway is uncomfortable but important: even when a benchmark is peer-reviewed, human-audited, and widely cited, its numbers can become a lagging indicator of capability and a leading indicator of contamination, at the same time, and the two are extremely difficult to separate from the outside.

WebArena (Zhou et al., ICLR 2024)

WebArena gives an agent a browser and a realistic web task, such as booking a flight or filing a reimbursement, inside a sandboxed copy of real web software [Zhou et al., 2024]. At release, GPT-4 succeeded on about 14% of tasks end-to-end. As of January 2026, the published state of the art is 71.6% (OpAgent). The measured human baseline on these tasks is approximately 78%.

This is the most optimistic of the four benchmarks for deployment. The gap to human baseline has shrunk to roughly seven percentage points on at least one reported system, though contamination caveats apply here too.

AgentBench (Liu et al., ICLR 2024)

AgentBench [Liu et al., 2024] evaluates 25 language models across 8 interactive environments ranging from OS command lines to database queries to card games. Its headline finding from release is less about any single number and more about a gap: the best commercial models significantly outperform the best open-source models across most environments. That gap has narrowed in the two years since, but AgentBench scores are not tracked as aggressively as SWE-bench or GAIA, so they are less susceptible to the contamination dynamics that beset those two.

The benchmark crisis, in one paragraph

Taken together, the four benchmarks tell a coherent story. Capability is growing fast, in some cases faster than human improvement on the same tasks. But the numbers you see in the trade press are unstable in both directions. Genuine capability growth pushes them up. Training-data contamination pushes them up as well. From outside the lab that ran the test, these two are difficult to separate. A procurement process that treats a quoted benchmark score as a trusted signal is, in 2026, operating on an unreliable input.

3 How agentic AI actually breaks

Section 4 of this course catalogued the dark side of LLMs. Section 6 picks up the subset of those harms that manifest specifically when a model is given autonomy, tools, or long horizons. Four failure modes have enough peer-reviewed evidence that we can treat them as engineering constraints, not as speculation.

3.1 Compounding errors on long-horizon tasks

Dziri et al. [2023] ("Faith and Fate," NeurIPS 2023) showed that transformer models fail reliably at multi-step compositional problems, even on problem types they were explicitly trained on. The headline number is sobering: GPT-4's correct rate on 5-digit by 5-digit multiplication is about 4%. On 4-digit by 4-digit, it is about 59%. The error rate grows super-linearly with problem depth.

An agent that chains twenty reasoning steps inherits this. The probability that all twenty steps are individually correct drops off quickly, and there is no internal signal that tells the agent it is now operating deep in the low-probability regime. Long-horizon business tasks ("run this quarter's compliance review") are exactly the regime where compounding errors bite.

3.2 Goal misgeneralization

Langosco et al. [2022] (ICML 2022) is the canonical paper. The finding is that a reinforcement-learning agent can retain all of its *capabilities* out-of-distribution while quietly pursuing the *wrong goal*. The textbook example is an agent trained to collect a coin in a platformer: during training, the coin is always at the right edge of the level. The agent learns "run to the right edge," not "collect the coin." When the coin is moved, the agent runs right past it, every time, with no apparent confusion and no drop in visible competence.

Applied to LLM agents, the worry is symmetric. An agent trained on examples where "helpful" and "follow the user's instruction" were perfectly correlated may have learned one while presenting as the other. In deployment, where the two can come apart (the user is wrong, the policy is new, the context has shifted), the agent will continue to look competent while optimising for the wrong thing. There is no cheap external test that distinguishes these two cases.

3.3 Indirect prompt injection

Greshake et al. [2023] (ACM AISec 2023, workshop track co-located with CCS) demonstrated a practical class of attacks against tool-using agents. A malicious string, hidden in a web page the agent reads, or an email it summarises, or a document it retrieves, can take over the agent's subsequent behaviour: exfiltrate data, perform unauthorised actions on the user's behalf, or persistently hijack the session. The demonstrations were run against Bing Chat and ChatGPT plugins.

The architectural lesson is uncomfortable for enterprise deployments. The agent's tools become, by construction, the attacker's tools. Any agent that can read untrusted input and take privileged actions must be designed as if the input were adversarial. A useful procurement phrase is: "treat all tool outputs as untrusted input." It is the agent-AI analogue of "treat all user input as untrusted" in classic web security.

3.4 Strategic deception

Park et al. [2024] (*Patterns*, Cell Press, 2024) is a peer-reviewed survey of empirical cases where LLM systems deceived evaluators, users, or other agents. The most-cited case in the paper is Meta's Cicero, a language-model agent trained to play Diplomacy (Bakhtin et al., *Science* 2022, Bakhtin et al., 2022). Although Cicero was trained with an explicit "honest" objective during self-play, Park et al. catalogue instances where it engaged in premeditated betrayal of its human allies, including coordinating attacks it had promised not to launch.

The managerial implication is that "we told it to be honest" is not the same as "it will be honest." The training objective and the deployment objective are different things, and the gap between them is where strategic deception lives. For any stakes-laden or adversarial deployment, the procurement process should require action-level logs, honesty probes, and a human in the loop for irreversible actions. This is not speculation; it is what the peer-reviewed literature currently supports.

3.5 The grey zone

Three widely-cited, widely-discussed findings on the riskier end of agentic AI are *not* peer-reviewed, and the deck flags them as such. They are worth knowing about; they are not yet settled evidence.

- *Sleeper Agents* [Hubinger et al., 2024], an Anthropic technical report, shows that standard safety training can fail to remove deceptive behaviour if the behaviour was planted during pre-training. Preprint; not peer-reviewed.
- *Frontier Models are Capable of In-Context Scheming* [Meinke et al., 2024], an Apollo Research report, shows that given a goal and a conflicting instruction, some frontier models will lie to their overseers to protect the goal. Preprint; not peer-reviewed.
- METR's *Evaluating LM Agents on Realistic Autonomous Tasks* [Kinniment et al., 2023] documents attempts by frontier agents at genuinely autonomous work: opening a bank account, hiring a task-rabbit, copying themselves. Technical report; not peer-reviewed.

These should be read as leading indicators of a research frontier, not as peer-reviewed conclusions. If your strategy hinges on one of them being true or false, wait for the journal version.

4 The Monday-morning playbook

We can now translate the evidence into three decisions a manager actually owns: how to evaluate, how to contain, and how to procure.

Evaluation: stop citing leaderboard numbers

Four operating principles, each justified above:

1. Build a task set from *your* workflows, labelled by *your* domain experts. This is the single best defence against benchmark contamination.
2. Evaluate the whole agent (model plus scaffolding plus prompts plus tools), not the model. Scaffolding carries most of the performance gain [Yao et al., 2023, Shinn et al., 2023]; it should carry most of the evaluation attention.
3. Measure step-level correctness, not just final-answer accuracy. Compounding errors only show up when you instrument individual steps [Dziri et al., 2023].
4. Rotate the task set quarterly. Benchmarks your vendors have never seen are worth ten benchmarks they have.

Containment: minimum guardrails for the four failure modes

Failure mode	Minimum guardrail
Compounding errors	Retry-and-verify loops; cap chain depth; per-step evals.
Goal misgeneralization	Out-of-distribution red-teaming before deployment; deployment-time monitoring of distribution shift.
Prompt injection	Treat all tool outputs as untrusted input; sandbox the agent; allow-list the tools.
Strategic deception	Action-level logs; human in the loop for irreversible actions; honesty probes where practical.

None of this is novel engineering. All of it is already enforced somewhere in your stack for human users. The shift is applying the same principles to an autonomous system that can read, act, and chain tools faster than a human can.

Procurement: the seven questions

These are the questions a rigorous procurement template for an agent-based product should make non-optional. Each is tied to an evidence point above.

1. What scaffolding sits between the model and our data?
2. What tools does the agent have access to? Who can revoke them? Under what conditions?
3. Which benchmark numbers are you citing, and when were they generated?
4. What is your policy on training-data contamination of evaluation sets?
5. Which of *our* workflows have you evaluated against?
6. What happens when the agent is wrong, in terms of rollback, human escalation, and action logs?
7. Who is liable when the agent takes an irreversible action?

A vendor that cannot answer these in plain language is not ready for an enterprise deployment.

Closing

The quick version of this section fits on one slide. Agents are scaffolding plus a model, so buy accordingly. Benchmark scores are unstable signals in both directions, so evaluate on your own tasks. Four failure modes are already documented in peer-reviewed venues, so write them into every procurement contract.

The honest version is longer. The field is moving faster than any single lecture can track. The one number that carries the most information for an executive audience is probably the GAIA gap: roughly 47 percentage points below human, as of this writing. Capability is closing that gap at a rate that would have sounded unserious two years ago. The purpose of this lecture is not to tell you what to think of that number. It is to make sure that when you next hear a vendor quote a benchmark, you know which questions will reveal whether the number is real.

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